

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Matrix Multiplication		
Student Name:	Manoj M	Reg. No.:	192511060
Section / Batch:	CSE	Date:	10.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Matrix Multiplication	Q1

SECTION A – CASE STUDY

QN)3D Graphics Rendering System (Matrix Multiplication) A graphics engine performs transformations using large matrices for rendering 3D objects

Analyze how divide and conquer matrix multiplication improves performance. Identify computational challenges in rendering. Apply optimized techniques such as Strassen's method. Analyze complexity and justify the solution.

Code of Conduct

I, Manoj M(192511060) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10%	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Algorithm – Strassen's Matrix Multiplication

1. **Start** and read two matrices A and B.
2. **Divide** each matrix into four equal submatrices.
3. **Recursively calculate** the seven Strassen products P1 to P7.
4. **Combine** these seven products to form the four parts of result matrix C.
5. **Return and display** the final matrix $C = A \times B$; stop.

Time Complexity: $O(n^{2.807})$

Advantage: Reduces matrix multiplication operations compared with the conventional $O(n^3)$ method.

Code:

```
2 {
3     int i, j;
4
5     for (i = 0; i < 2; i++)
6     {
7         for (j = 0; j < 2; j++)
8         {
9             C[i][j] = A[i][j] + B[i][j];
10        }
11    }
12 }
13
14 void sub(int A[2][2], int B[2][2], int C[2][2])
15 {
16     int i, j;
17
18     for (i = 0; i < 2; i++)
19     {
20         for (j = 0; j < 2; j++)
21         {
22             C[i][j] = A[i][j] - B[i][j];
23        }
24    }
25 }
26
27 void multiply(int A[2][2], int B[2][2], int C[2][2])
28 {
29     int i, j, k;
30
31     for (i = 0; i < 2; i++)
32     {
33         for (j = 0; j < 2; j++)
34         {
35             C[i][j] = 0;
36
37             for (k = 0; k < 2; k++)
```

Output

```
Enter Matrix A (4x4):  
Enter Matrix B (4x4):  
  
Result Matrix using Strassen's  
Method:  
12 8 8 12
```